

Task-specific recovery and robustness in computational chart selection

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Abstract

Selecting a chart by a numerical score requires evidence that the ranking survives reasonable changes in how the chart is read and displayed. We develop a computational benchmark that links a declared two-sample task to rendered pixels, recovery error and selection robustness. The task is to recover five empirical quantile differences. Nine implemented chart families provide 23 settings, including empirical cumulative distribution and direct quantile baselines. Across 127 comparisons from nine public medical and digital-health resources, the reference median best score is 99.172 on a scale approaching 100 as normalized error approaches zero. A fixed five-quantile plot outperforms the best of the original seven families in 124 comparisons. Winning families change in 13 comparisons between 128 and 512 pixels, and 51 comparisons admit more than one family within an excess-error budget of 0.01 pooled standard deviations. We supplement reference rankings with group-specific recovery error, equal tuning budgets, resource-disjoint retrospective evaluation, 72 synthetic stress tests and minimax-regret selection tested at a new display resolution. The contribution is an executable audit of task-specific chart choices and their limits. Two uncalibrated computational observers do not establish human reading accuracy; prospective reader validation remains necessary for that claim.

Introduction

A chart should be evaluated against the question its reader is expected to answer. Comparing two means, comparing distributional shapes and estimating individual change are different tasks. Displays that omit the required information cannot become adequate simply through higher resolution. Conversely, retaining observations does not establish that a reader can recover the relevant information accurately. Distributional comparisons and shift functions have long provided alternatives to mean-only summaries.^{1;2} Scientific visualization requires explicit objectives and evidence that the resulting displays support those objectives.^{3;4}

Automated graphical design, constraint-based systems and learned recommenders address chart construction and recommendation.⁵⁻⁷ Task taxonomies clarify the analytic operations a visualization should support, while graphical-perception experiments evaluate human judgments.^{8;9} These contributions motivate, but do not validate, a separate computational question: given known data and a specified recovery algorithm, which rendering most accurately communicates a declared numerical target?

A score alone does not answer whether the recommended chart is useful. A color-calibrated decoder may recover faint heatmap intensities more accurately than a person. A mean-only baseline can be predictably weak for a distributional task. Searching more settings in one family can increase its opportunity to win. Nearly identical scores may exchange rank with a small change in resolution. These concerns require explicit controls rather than a transformation that forces every chart to have a unique score.

We address this problem through a benchmark with three outputs: task-specific recovery, group-specific fidelity, and sensitivity of the selected rendering. We add strong distributional baselines to seven familiar chart families, retain failures and genuine ties, and evaluate both adaptively optimized and fixed settings. A minimax-regret rule provides a complementary choice when one setting must

work across declared computational environments. We test that choice on a display resolution excluded from its selection. All analyses remain conditional on their renderers, tasks and observers.

Results

A declared task connects the image to the score

For samples x and y , the evaluator computes

$$\Delta_k = Q_y(p_k) - Q_x(p_k), \quad p_k \in \{0.10, 0.25, 0.50, 0.75, 0.90\}. \quad (1)$$

These five equally weighted differences describe the lower, middle and upper central portions of two marginal distributions. They do not estimate diagnostic accuracy, a causal treatment effect or within-person change. The empirical quantiles use linear interpolation.

Let c identify a chart family and setting, and e a resolution–observer environment. The renderer produces an image $I_{c,e,r}$ at placement r . The decoder receives only that image and public calibration metadata A_c , including chart type, panel bounds and measurement limits. It returns estimated group quantiles $\tilde{Q}_{g,c,e,r}(p_k)$. Observations, target quantiles and stored summary values are unavailable to the decoder. Define

$$L_{c,e,r} = \frac{1}{5s} \sum_{k=1}^5 |\tilde{Q}_{y,c,e,r}(p_k) - \tilde{Q}_{x,c,e,r}(p_k) - \Delta_k|, \quad (2)$$

$$\bar{L}_{c,e} = \frac{1}{4} \sum_{r=1}^4 L_{c,e,r}, \quad S_{c,e} = \frac{100}{1 + \bar{L}_{c,e}}, \quad (3)$$

$$s^2 = \frac{(n_x - 1)s_x^2 + (n_y - 1)s_y^2}{n_x + n_y - 2}. \quad (4)$$

All candidates for a comparison share the same pooled within-group scale s . A recorded one-source-unit fallback is retained for zero scale. This special case is not unit invariant. A score of 80 means an average absolute contrast error of $0.25s$, not 80% correct readers or retained information.

Since $dS/dL = -100/(1 + L)^2 < 0$, maximizing S is exactly equivalent to minimizing the declared loss. The reciprocal transformation adds interpretability, not information or mathematical novelty. Equal losses remain equal after any strictly monotone transformation. We retain computational ties within absolute loss tolerance 10^{-12} and report winning margins rather than manufacturing uniqueness.

Strong baselines expose the value of task-matched displays

The expanded benchmark evaluates 23 settings across nine families (Table 1). The original seven families comprise mean bars, dot plots, box plots, variability intervals, histograms, probability heatmaps and violins. Two new baselines address the distributional task directly: an empirical cumulative distribution function (ECDF) and five quantile markers. The latter shows exactly the group summaries needed for the target; it is a deliberately strong task-matched control, not a novel visualization. The renderer may use data to construct the chart, but the decoder must recover its values from pixels. This distinction prevents target leakage into recovery while allowing task-appropriate encodings.

The primary arrays contain 127 feature comparisons from nine public resources.^{10–18} The reference condition is the original threshold observer at 256 pixels per group panel, averaged over four placements. The new implementation reproduces every original reference-setting loss, including failures, for the 21 existing settings. This parity check isolates the added comparisons from changes to the historical calculation.

A chart score needs a task and a robustness check



Figure 1: **An explicit contract for computational chart selection.** The target, calibration and recovery algorithm are declared before evaluating the rendered pixels. Reference recovery and worst-environment regret answer different questions. The displayed tumour-radius-standard-error example uses actual data and computed scores; its ranking includes all nine evaluated families. The direct quantile plot uses percentile on its horizontal axis. No score represents measured human performance.

Table 1: **Candidate settings and the equal-budget control.** Fixed settings are chosen before the expanded results are inspected. The original 15-family contextual catalogue is retained in the Supplement; it is not a claim that all 15 support this task.

Family	Full search	One-setting control
Bar chart	Group means	Group means
Dot plot	Mean dots; opaque raw dots	Raw dots
Box plot	Minimum, quartiles, median, maximum	Same
Interval plot	Five-number variability display	Same
Histogram	4, 8, 16, 32, 64 bins	16 bins
Heatmap	4, 8, 16, 32, 64 probability bins	16 bins
Violin plot	8, 16, 32, 64, 128, 256 density grid points	64 points
ECDF	Full empirical cumulative step function	Same
Quantile plot	Five target group quantiles	Same

The median best reference score is 99.172. The fixed quantile and ECDF baselines have median scores 99.167 and 97.826, respectively; the median best score among the original seven families is 96.414. The quantile baseline beats that seven-family optimum in 124 of 127 comparisons, and the ECDF does so in 76. Table 2 reports adaptive family scores and wins under an equal setting budget. Figure 2 shows the distribution of results, rather than only favorable examples. Family wins are descriptive development outcomes, not independent replications of superiority.

Table 2: **Reference results across 127 comparisons.** “Wins” permits all 23 settings; “Fixed wins” permits one setting per family. Median S and group error G use each family’s best valid reference setting. Failed settings are excluded from selection, not assigned favorable values.

Family	Wins	Fixed wins	Median S	Median G	Valid
Quantile plot	122	123	99.17	0.014	127
Heatmap	2	1	96.18	0.046	127
ECDF	2	2	97.83	0.023	127
Violin plot	1	0	92.91	0.131	127
Interval plot	0	0	69.94	0.260	127
Histogram	0	0	94.98	0.057	127
Dot plot	0	1	95.58	0.040	127
Box plot	0	0	70.59	0.260	119
Bar chart	0	0	75.10	0.671	122

Changing the requested information tests task dependence

A secondary analysis keeps the nine fixed chart settings unchanged and changes only the requested target: the median contrast, the original five contrasts, or 19 contrasts at probabilities 0.05, 0.10, $\dots$, 0.95. The five-marker quantile display is not redrawn to contain 19 marks: its recovered values are interpolated between markers and extrapolated constantly outside the plotted probability range. Other distributional decoders invert their recovered distributions at the requested probabilities. This secondary analysis was motivated by the observed dominance of the task-matched baseline and was specified before its own execution; it is hypothesis-generating rather than an independent confirmatory test.

Relative to the five-quantile task under the same fixed budget, the threshold-observer winning family changes in 112 comparisons for the median-only task and 120 for the 19-quantile task (Fig. 2d). The task-specific counts and all losses are supplied. For the median-only task, interval plots belong to the threshold observer’s winning set in 70 cases but the alpha observer’s winning set in 14; quantile plots belong to those sets in 23 and 80 cases, respectively. Tied memberships can exceed the number

Strong baselines test what chart optimization adds

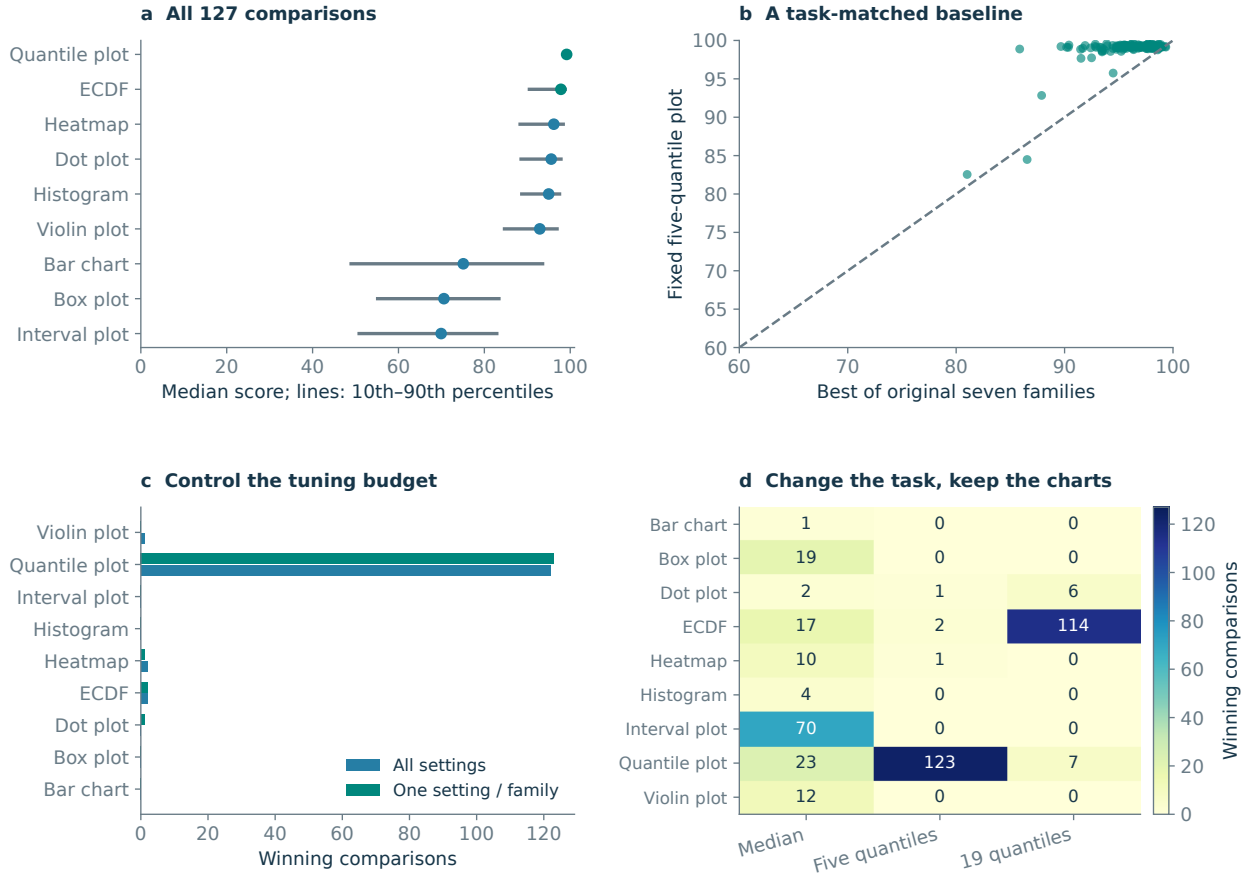


Figure 2: **Comparisons against strong and equal-budget baselines.** **a**, Distribution of reference family scores; intervals show descriptive 10th–90th percentiles across features, not confidence intervals. **b**, Fixed quantile plots versus the best of the original seven families; points above the diagonal favor the quantile baseline. **c**, Winning-family counts with adaptive settings and one fixed setting per family. **d**, Fixed-budget winning memberships (including ties) when the target changes from the median to five or 19 quantile differences while the rendered charts remain unchanged. The task-sensitivity analysis was added retrospectively after observing the strong quantile-baseline results. These panels test computational recovery, not reader preference.

of comparisons. This observer dependence prevents treating the median-task ranking as a human recommendation. These comparisons make the scope of a winner concrete: changing the information requested can change the preferred encoding even when the data, panel size and candidate renderings remain fixed.

Group fidelity and trivial baselines qualify a high score

Contrast recovery can hide errors in both groups. We therefore report, without combining it into a new weighted score,

$$G_{c,e} = \frac{1}{4} \sum_r \frac{1}{10s} \sum_{g \in \{x,y\}} \sum_k |\tilde{Q}_{g,c,e,r}(p_k) - Q_g(p_k)|. \quad (5)$$

By the triangle inequality, $\bar{L}_{c,e} \leq 2G_{c,e}$, but a small L does not imply a small G . Adding the same error to both recovered groups yields zero contrast error while distorting both distributions. This is a mathematical limitation of the objective, not a decoder-specific anomaly.

A second control predicts zero difference at every quantile, with $L_0 = (5s)^{-1} \sum_k |\Delta_k|$. We report $1 - \bar{L}/L_0$ when $L_0 > 0$; negative values mean that decoding performs worse than this trivial prediction. When $L_0 = 0$, the relative skill is undefined and remains missing. Full group errors, zero-baseline losses and relative skills accompany every family result. Near-null contrasts can have high absolute scores even when the image adds little beyond predicting no difference.

A more basic limit occurs before rasterization: different distributions can have the same displayed summary. Under a common fixed axis, two mean-only displays may be identical despite different quantile targets. No decoder receiving the same image and metadata can distinguish those cases perfectly. The Supplement gives an explicit equal-mean, equal-variance construction and a lower bound. This motivates strong baselines that actually encode the declared target.

Robust choices complement precise-looking reference ranks

The best family changes in 13 of 127 comparisons between 128 and 512 pixels. The alternative alpha-based observer changes 14 reference winners at 256 pixels. These observers share chart geometry and design assumptions; their comparison is algorithmic sensitivity, not independent perceptual validation. Tight measurement axes are evaluated separately, excluding mean bars from both sides because removing zero can invalidate their baseline.

The median best-versus-second-best family score margin is 1.272; 2 comparisons have margins below 0.1 points. Reporting three decimals therefore does not imply that the differences are perceptible or stable. For an explicit excess-error budget ϵ , define

$$\mathcal{C}_\epsilon(e) = \{c : \bar{L}_{c,e} \leq \min_j \bar{L}_{j,e} + \epsilon\}. \quad (6)$$

We report family membership at $\epsilon \in \{0.001, 0.005, 0.01, 0.025\}$ pooled standard deviations. At $\epsilon = 0.01$, 51 comparisons admit multiple families. These are declared computational tolerances, not empirically validated thresholds of human equivalence.

If a single unchanged setting must operate across environments $\mathcal{E}$, define

$$R_c = \max_{e \in \mathcal{E}} \left[\bar{L}_{c,e} - \min_{j \in \mathcal{C}_{\text{valid}}(e)} \bar{L}_{j,e} \right], \quad \mathcal{C}_{\text{robust}} = \arg \min_{c \in \cap_e \mathcal{C}_{\text{valid}}(e)} R_c. \quad (7)$$

The six selection environments combine two observers with three resolutions. A candidate must decode at every placement in every environment to qualify as robust. The environment-specific comparator may use any setting valid in that environment. This is a standard minimax-regret construction applied to a declared chart benchmark; it is not a claim of a new optimization theorem.

Its training-envelope advantage follows from its definition and must not be mistaken for out-of-environment validation.

We therefore freeze each comparison’s reference and minimax settings and transfer them to 384-pixel panels with placements 0.125, 0.375, 0.625, 0.875, which are excluded from selection. The fixed quantile plot is transferred alongside them. Figure 3 and the Supplement report resource-balanced errors for both observers, including any failure. At the new display condition, resource-balanced mean losses for the fixed quantile, reference-selected and minimax-selected charts are 0.00638, 0.00646 and 0.00655 for the threshold observer, and 0.00339, 0.00383 and 0.00351 for the alpha observer. Thus minimax selection does not consistently improve on reference selection, and the simple fixed quantile baseline has the lowest mean loss for both observers. This separates retrospective optimization from performance at a new display condition. It does not test an unseen medical population.

Resource transfer and simulation extend the development audit

A leave-one-resource-out analysis selects one global configuration using mean loss on the other eight resources, weighting resources equally. The selected setting is then evaluated on every comparison in the omitted resource without case-specific tuning. Its resource-balanced held-out loss is 0.0093. The chosen configurations and comparison-level outputs are reported in the Supplement. Because all resources were available during development, this is retrospective resource-disjoint evaluation, not untouched external validation.

Seventy-two synthetic comparisons span null, location-shift, scale-change, skewed, mixture and heavy-tailed regimes; sample sizes are 20, 100 and 500 per group, with four replicates per cell. The nine fixed settings, seeds and generation rules are declared before the synthetic run. Scores are evaluated against each generated sample’s empirical target, not its population quantile difference. These modest simulations expose controlled failure modes; they do not establish universal performance across distributions or replace human observations.

Data-specific sensitivities omit the two flagged CKD potassium values without replacing them and repeat results after excluding the 16 retinopathy features with unresolved participant linkage. The Parkinsons comparisons remain outside the primary analysis because parsed identifier groups conflict with documented participant counts. No data correction is chosen to improve chart scores.

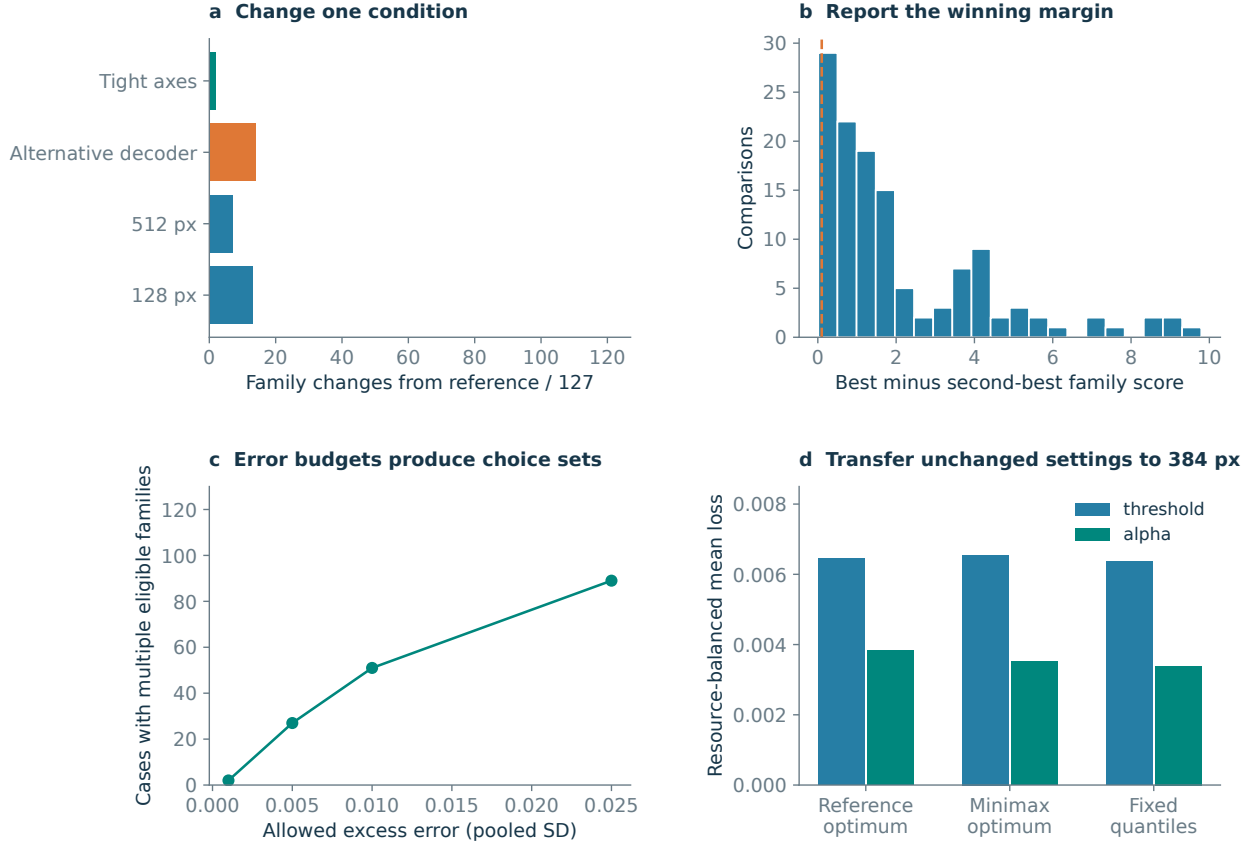
Figure 4 retains three examples from the earlier manuscript: two substantial mean-bar gains and one high-scoring baseline. All nine-family rankings are recalculated. An expanded candidate set can change the earlier winner; the examples are not treated as representative estimates of benefit. A small incremental score gain is a descriptive computational result and does not imply statistical equivalence or a ceiling on human performance.

Discussion

The benchmark makes chart selection inspectable by specifying what is to be recovered, what information reaches the decoder, how loss is normalized and how stable the selected setting remains. Its main contribution is the combination of strong task-matched baselines, explicit decoder dependence, separate group-fidelity auditing and transfer of unchanged settings. The numerical score is useful only within that declared contract.

The added quantile and ECDF baselines are substantive tests. Beating a mean-only display on a distributional question is insufficient evidence that adaptive chart selection adds value. A direct quantile display may be an economical answer when the requested information is exactly five quantiles. Its success would support matching the encoding to the task, rather than demonstrate a universally optimal chart. Other objectives, including temporal dynamics, association, uncertainty, spatial

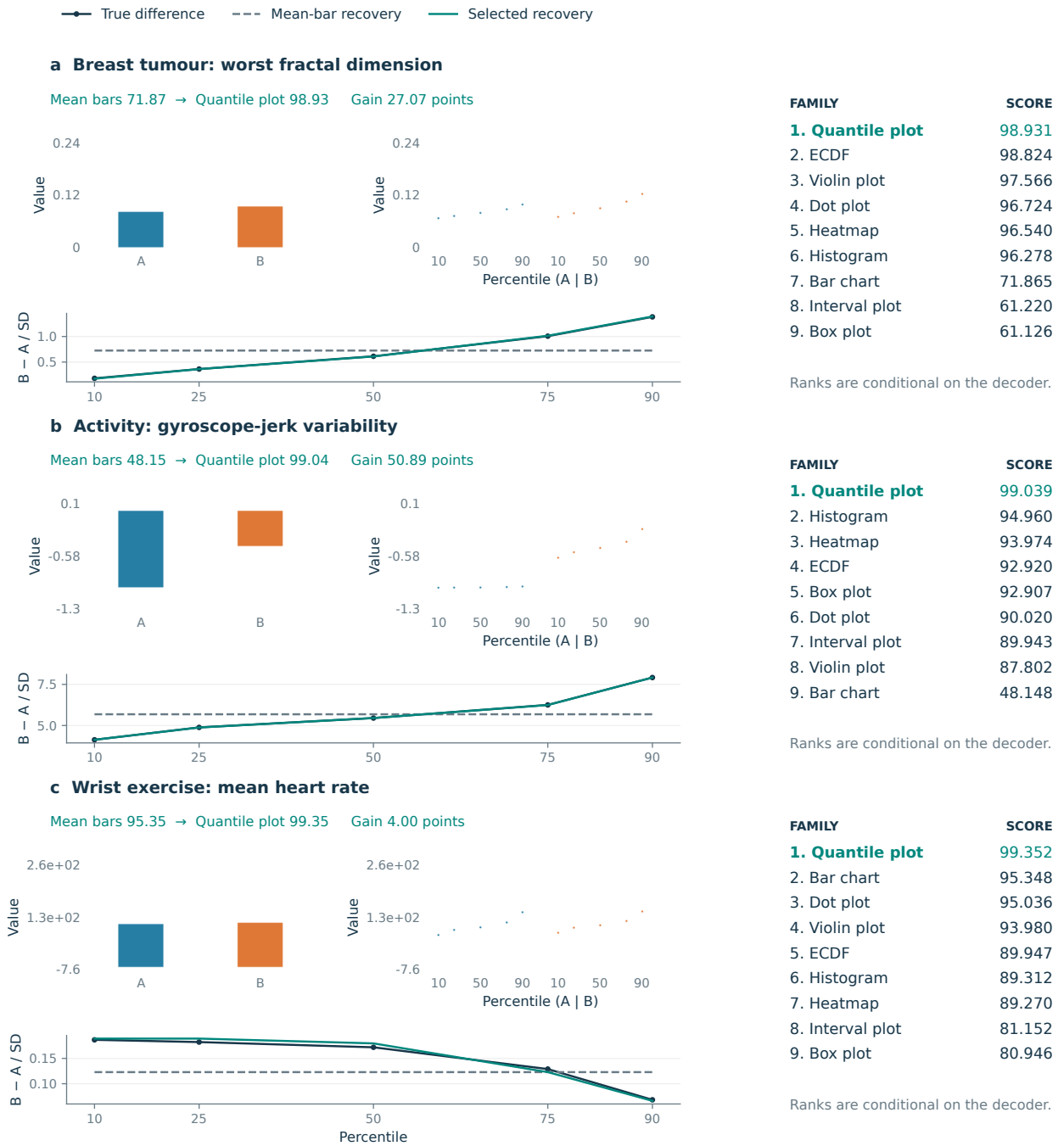
A numerical winner can be fragile



Error budgets and display perturbations are computational checks, not confidence intervals or human-equivalence thresholds.

Figure 3: **Selection stability and transfer.** **a**, Changes in the winning family relative to the reference condition; the tight-axis comparison excludes bars on both sides. **b**, Reference winner–runner-up score margins. **c**, Comparisons admitting multiple families as the allowed excess loss increases. **d**, Unchanged selected settings tested at a new resolution and placements; means first average within resources and then weight the nine resources equally. Minimax training-envelope performance is not presented as evidence of external generalization.

What the selected chart recovers—and what it adds



Examples are retained from the earlier revision, not selected afresh for the largest gains. No reader benefit is inferred.

Figure 4: **Large and limited gains over mean-only displays.** The breast-tumour, activity and wrist examples are retained from the earlier revision.^{10;16;18} Left: mean bars; middle: the best current reference rendering; right: all nine family scores. Lower panels compare the empirical quantile contrast with recovered contrasts, averaged over the four placements. Scores average absolute losses before transformation. The plotted line error therefore need not equal the scored error. A high baseline score can leave little room for improvement.

pattern or rare-tail events, require different targets and candidate sets. The 15 familiar chart names in the contextual catalogue do not make those tasks interchangeable.

Decoder dependence remains a central limitation. The threshold observer and alpha observer use known calibration and family-specific recovery rules. Neither is independently trained or validated against human behavior. A probability heatmap’s calibrated intensities are machine-readable without establishing perceptual accessibility.¹⁹ Opaque-dot overlap can impair an area-based decoder without implying that a different algorithm or person fails identically. Agreement between the two observers is therefore reassuring only within this narrow class of algorithms. Disagreement reveals fragility; it does not identify which observer better models a reader.

The benchmark also separates three kinds of uncertainty. Placement and resolution checks address rendering sensitivity. Resampling resource-level summaries describes variation among the nine convenience resources. Sampling uncertainty in population quantile differences is a different problem and is not estimated by either procedure. Participant-level source aggregation reduces repeated-measure inflation in HAR, BIDMC and wrist data, but features within a resource remain correlated. Small wrist cohorts, unresolved retinopathy linkage and questionable CKD values constrain biological interpretation. We study empirical image recovery, not disease effects inferred from these resources.

The zero-inclusive axis and SD normalization are deliberate but imperfect choices. Including zero can waste space for non-bar encodings; tight-axis results expose this dependence. Pooled SD is sensitive to extremes. Changing to another positive scale shared across candidates would rescale their losses without changing the ranking for that comparison; it could change pooled summaries across comparisons. Zero-scale fallback remains source-unit dependent. We disclose these limits rather than presenting a score threshold as a universal quality certificate.

Human validation remains the decisive next step for claims about readers. A prospective experiment must freeze stimuli and computational predictions, use untouched cases, randomize display assignment, and measure estimation error alongside time and confidence. It should test whether the score adds predictive value beyond chart family and direct quantile baselines, accounting for repeated participants and cases. The accompanying protocol specifies these analyses and provides planning tools, but supplies no recruited participants, responses or approval claims. The present evidence supports computationally auditable choices for a fixed task; it does not establish improved human interpretation or clinical decisions.

Methods

Source arrays and units of analysis

The original prepared arrays are retained: WDBC 30 comparisons, Cleveland five, heart failure six, CKD 11, ILPD nine, retinopathy 16, HAR 40, BIDMC five and wrist five. HAR uses subject/activity medians from 30 participants. BIDMC summarizes readings within 46 source patients. Wrist exercise uses early/late ECG-reference summaries within eight participants.^{20;21} Missing observations and transformations remain recorded in the preparation pipeline. Paired resources are explicitly labeled, but the target compares marginal quantiles. Parkinsons contributes 22 exploratory comparisons only in the archived analysis.

Rendering and recovery

The common reference interval includes zero and extends four times the larger group Scott bandwidth beyond the observed extremes, with a bandwidth floor of 0.01s. Each group occupies a separate square panel of side 128, 256 or 512 pixels, with common measurement limits. Images are drawn at fourfold resolution, reduced with Lanczos filtering and blurred by a 0.5-pixel Gaussian. Original mark sizes and colors are unchanged. Four vertical placements are evaluated; raw-dot jitter also

changes deterministically, whereas heatmap placements repeat the same strip. These are specified stress conditions, not independent random reader samples.

The original observer extracts thresholded colored or black pixels. The alternative projects RGB deviation from white onto the known group-color direction, clips the resulting alpha estimate to $[0, 1]$, uses an alpha threshold of 0.2 for geometric masks and integrates alpha above 0.05 for area recovery. Heatmap recovery averages a local 3×3 neighborhood in the alternative observer. Both observers retain known layout and family-specific geometric assumptions. The Supplement and source code give exact rules and failure conditions.

The ECDF is a two-pixel step curve with cumulative probability on the horizontal axis and measurement on the vertical axis. Decoding estimates the ordinate at each target probability column. Finite-sample ECDF inversion differs from linear-interpolated empirical quantiles, a retained source of error. Quantile plots place radius-two-pixel markers at the five target probabilities; decoding uses local pixel-mass centroids, with no numerical target values in metadata. These baselines occupy the same panel area as the other charts.

Selection and validation

Loss is averaged across four placements before transformation. A failed placement invalidates the setting in that environment. Family summaries use the minimum valid loss within family. Exact ties use absolute tolerance 10^{-12} ; deterministic display order does not imply scientific superiority. Equal-budget controls use the fixed settings in Table 1. The minimax rule requires a configuration to succeed in all six selection environments. The new-resolution transfer uses fixed selections and new placements without retuning.

Resource-disjoint selection requires a candidate to decode every training comparison, averages feature loss within each training resource and then weights resources equally. The selected configuration's held-out failures are retained. Descriptive bootstrap intervals for each family's loss difference from the quantile baseline resample resource means 5,000 times with a fixed seed; invalid comparisons are omitted and counts disclosed. These intervals concern the sampled resource summaries and are not population-generalization guarantees.

The protocol is a retrospective extension, not a preregistration. A deterministic fixture failure motivated increasing the new ECDF stroke from one to two pixels before the final run; the stopped initial run supplies no reported results. Protocol, source and input hashes are recorded. Checks cover all 23 settings and both observers, including blank-image rejection, pixel dependence, positive unit rescaling and metadata restrictions. Original reference losses are independently compared with the supplied archived CSVs. Source code, per-trial recovered values, aggregate outputs and figure-generation scripts are included.

Data availability

The source resources are available through the cited UCI and PhysioNet records. The accompanying reproducibility archive contains prepared arrays, source copies where supplied, acquisition records, attribution, transformations and checksums. Source-specific terms are documented in `DATA_PROVENANCE.md`; original licenses are retained. Synthetic inputs are explicitly identified and included. No human-reader data were collected.

Code availability

The complete version 2 analysis accompanies this manuscript in the reproducibility archive. Running `python reproduce_strengthened.py` executes numerical checks, the expanded benchmark, sen-

sitivity and transfer analyses, and figure generation. Pinned dependencies, deterministic seeds, source hashes and per-trial CSVs are supplied. The associated project repository is <https://github.com/Elgendi/two-sample-chart-selection>; the accompanying archive identifies the exact code underlying this version.

Author contributions

M.E. conceived the study, developed the methodology, performed the analyses, interpreted the results, and wrote and revised the manuscript.

Competing interests

M.E. serves on the editorial boards of Communications Medicine and npj Biosensing. The author declares no other competing interests.

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